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SPECIAL ISSUE EU Space Governance at the Threshold of a New Era

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ABSTRACT

This introductory article to the special issue "EU Space Governance at the Threshold of a New Era" explores the European Union's evolving role as a strategic actor in space. It argues that the EU is undergoing a fundamental shift from a regulatory to a security-oriented approach, driven by institutional consolidation, technological ambition, and geopolitical pressures—particularly following Russia's 2022 invasion of Ukraine. The article is divided into two parts. The first outlines key conceptual and institutional developments in EU space governance, including the emergence of hybrid governance models, the securitization of flagship programmes, and the launch of new initiatives such as IRIS^2. The second part introduces the seven contributions to the special issue, which examine core themes such as dual-use infrastructure, cybersecurity, environmental sustainability, industrial policy, and legal harmonization. Together, these articles provide a comprehensive and critical assessment of the EU's transition toward strategic autonomy in space. They highlight both the opportunities and constraints of this moment, offering insights into the governance choices that will shape Europe's space future.

1 | Conceptualizing Europe's Strategic Repositioning in Space

The European Union's ("EU" or "Union") involvement in space governance has reached a critical inflection point. Over the last two decades, the EU has cultivated a growing institutional and legal presence in the space domain, positioning itself not just as a contributor to global space governance, but increasingly as a strategic actor in its own right. This transformation has been shaped by the Union's evolving policy ambitions, technological capabilities, and geopolitical realities. Today, space is no longer conceived merely as a domain of scientific exploration or economic opportunity. It is now seen as an essential component of European sovereignty, security, and competitiveness.

This special issue of *Global Policy* builds on discussions held during the first International Conference on EU Space Governance, organized by the Leuven Centre for Global Governance Studies of KU Leuven, and hosted in Brussels on 22 May 2024. It examines the foundations and future trajectories of EU space governance. It takes stock of how the Union's role in space has evolved in response to endogenous institutional developments and exogenous geopolitical shocks, and how this evolution intersects with the broader pursuit of European strategic autonomy. We also identify the key conceptual and policy shifts that mark the EU's current transition from a regulatory power in space to a strategic space actor.

1.1 | From Coordination to Governance: Legal and Institutional Foundations

The EU's involvement in space policy has developed incrementally, beginning with support for research and industrial policy through the Union's Framework Programmes, and maturing into a fully-fledged legal and institutional architecture (European Parliament. Directorate General for Parliamentary Research Services. 2017; Pavesi and Wouters 2023; Wouters 2003). The

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2007 Lisbon Treaty granted the EU a shared—rather parallel (see European Commission and HR 2022)—competence in space affairs, formalizing its role and allowing for a more coherent policy approach (Articles 4(3) and 189 of the Treaty on the Functioning of the EU). This was institutionalized further through a dedicated Space Regulation in 2021, which established the Union Space Programme and created the European Union Agency for the Space Programme (EUSPA) (Regulation (EU) 2021/696, 2021). The Regulation consolidated existing flagship initiatives such as Galileo, Copernicus, and the Space Surveillance and Tracking (EUSST) initiative under a unified legal umbrella (Jacobs 2024b; Peldszus and Faucher 2022; Potter 2024).

While the Union Space Programme has introduced greater coherence, questions remain over the governance balance between supranational and intergovernmental actors. EUSPA, albeit formally empowered, has seen its role debated among stakeholders (Kosińska 2023). The European Space Agency (ESA), as an intergovernmental body with extensive technical expertise, continues to play a central role in programme implementation. The complexity of EU-ESA relations has led to a hybrid governance model, which some have called "functional complementarity", but others view as a source of inefficiency and institutional fragmentation (Lieberman and Hoerber 2024; Wouters and Pavesi 2022; Mazurelle et al. 2009).

This hybridism is not purely accidental. It reflects deeper tensions in the EU's institutional design: between efficiency and legitimacy, expertise and accountability, flexibility and coherence. These tensions are particularly salient in space, where decisions often implicate high-tech industrial policy, sensitive security concerns, and transnational legal coordination.

1.2 | Strategic Shock and the Geopoliticization of Space

Russia's full-scale invasion of Ukraine in February 2022 marked a watershed moment in EU space governance. It catalyzed a rapid reframing of space policy from a developmental and regulatory agenda to one driven by considerations of resilience, security, and strategic autonomy (Bombeek 2023; European Commission and HR 2025). Space systems such as Copernicus and Galileo, once promoted primarily for civilian and commercial use, were rapidly recognized as critical dual-use infrastructure, vital for both humanitarian response and defense operations.

In response, the EU launched the Secure Connectivity Initiative, laying the legal foundation for the IRIS^2 satellite constellation (Regulation (EU) 2023/588, 2023). This system is envisioned as a sovereign, secure, and multi-orbital communications infrastructure, strategically designed to reduce dependence on foreign providers and ensure secure communications in times of crisis. IRIS^2 exemplifies the Union's shift from space as a public good to space as a security enabler (Sigalas 2023).

The war in Ukraine also underscored Europe's vulnerability in the face of asymmetric threats and technological dependencies. The Ukrainian military's reliance on U.S.-based commercial services such as Starlink illuminated the EU's lag in autonomous capabilities. This has reinvigorated political calls to accelerate the Union's capacity for secure satellite-based services and has led to proposals for expanding Copernicus Security Services and reinforcing European access to satellite bandwidth (Cellerino 2023; Hollinger et al. 2025).

The March 2025 Joint White Paper for European Defense Readiness further institutionalizes this shift (European Commission and HR 2025). It identifies secure space-based systems—for communications, navigation, surveillance, and threat detection—as integral to EU deterrence and readiness. The document goes beyond rhetorical ambition and proposes concrete actions: integrating Ukraine into the EU Space Programme, prioritizing space infrastructure under dual-use and defense planning, and creating stockpiles and strategic reserves of critical space technologies.

1.3 | Recasting EU Space Governance: Strategic, Legal, and Institutional Dimensions

The EU's space strategy is now being recast as part of a broader transition toward a European Defense Union. The 2023 EU Space Strategy for Security and Defense frames space as a contested and congested domain, increasingly vulnerable to cyberattacks, anti-satellite weapons, and hybrid interference (European Commission and HR 2023). This strategy outlines operational and legal responses, including resilience-building, coordination with NATO, and steps toward a more robust EU Space Law.

The upcoming EU Space Law is expected to establish harmonized rules for space operations across the Union. Its goal is to strengthen the single market for space, enhance the safety and sustainability of space activities, and bolster the competitiveness of the European space industry (DEFIS 2024). While many stakeholders support this initiative, the legislative proposal has faced delays, and the timing and final scope of the Law remain uncertain. Several articles in this special issue engage directly with issues the EU Space Law may address, offering legal, policy, and industrial recommendations.

As a legal project, EU space governance is marked by incrementalism rather than codification. While the EU's 2021 Space Regulation provides a basis for programmatic governance, broader legal harmonization remains incomplete. Scholars have emphasized the need for a clearer articulation of liability, registration, environmental sustainability, and private sector regulation (Jacobs 2024a; Peldszus and Faucher 2022). Meanwhile, EU institutions have expressed interest in streamlining defense-related procurement and certification to facilitate dual-use technology integration (European Commission and HR 2025).

The EU's strategy also intersects with industrial policy. Space remains a cornerstone of the European defense industrial base, and funding mechanisms such as the European Defense Fund (EDF), the European Investment Bank's Security and Defense Action Plan, and the newly proposed SAFE instrument (Security and Action for Europe) are explicitly directed at strengthening space-industrial capabilities. The EDF in particular supports SMEs, startups, and transnational consortia that are increasingly

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central to innovation in quantum communication, AI-powered satellite systems, and cyber-resilience technologies (Lieberman and Hoerber 2024; Remuss 2018).

1.4 | Normative Ambition and Geopolitical Constraints

While the EU aspires to shape global space governance, it faces significant constraints. The geopolitical order is fragmenting, and multilateral fora such as the United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) are increasingly gridlocked. The EU's normative project—promoting space sustainability, peaceful use, and global commons governance—must now contend with a world of intensifying competition, where space capabilities are openly weaponized and commercial actors are reshaping access regimes (Kolovos 2023).

Moreover, the EU's internal cohesion remains fragile. Space policy reveals the uneven capacity of Member States to invest in infrastructure, harmonize procurement, and coordinate security priorities. The fragmentation of national regulatory environments also hampers market integration. While Permanent Structured Cooperation (PESCO) offers a legal tool for voluntary integration, its implementation in space projects remains limited.

Nevertheless, there are notable signs of momentum. The White Paper's call for a European Military Mobility Strategy includes provisions for integrating space systems into broader dual-use logistics (Chihaia 2024). Ukraine's potential participation in EUSPA and the Coordinated Annual Review on Defense (CARD) would mark a step toward transnational defense–space integration. In this regard, the EU is beginning to define a uniquely European model of space governance: decentralized, civilian-led, but strategically aligned.

1.5 | Conclusion: EU Space Governance at the Threshold of a New Era

EU space governance now stands at the threshold of a new era. The convergence of geopolitical risk, technological acceleration, and institutional maturity has created both opportunity and urgency. As the EU contemplates a Defense Union, space is no longer a peripheral concern; it is a central theater of strategic competition and a core component of resilience.

This first part of the introduction has traced the conceptual and institutional shifts in EU space policy, situating them within the broader dynamics of European integration and geopolitical realignment. The second part will offer a thematic synthesis of the contributions to this special issue, each of which explores a different aspect of this evolving governance landscape: from legal frameworks to industrial policy, from military-civil fusion to international diplomacy.

2 | Contributions to This Special Issue

This special issue brings together seven original articles that explore the multi-dimensional evolution of the Union's space governance. Each contribution addresses a distinct yet interconnected challenge, from EU constitutional law and cybersecurity to industrial strategy and environmental responsibility. Together, they provide a comprehensive and critical examination of the EU's pursuit of strategic autonomy and a coherent space governance in a rapidly transforming geopolitical environment.

2.1 | A Gradual Shift in EU Space Policy (Müller and De Man)

The opening article offers a longitudinal account of the EU's shift from a fragmented actor with limited competence in space to one pursuing an integrated regulatory and strategic agenda. Through institutional analysis and policy tracing, Müller and De Man show how legal ambiguity and multi-level governance gradually gave way to more assertive EU leadership, especially through instruments like the Union's 2001 Space Regulation. Yet, the article cautions against overstating coherence: deep tensions remain in EU-ESA relations, competence overlaps continue with Member States, and there is the uncertain future of EU Space Law. The article provides a vital historical anchor and highlights the enduring importance of legal design in shaping the EU's space ambitions.

2.2 | The Dual-Use Conundrum in EU Space Law (Bennett)

Bennett's contribution dissects the constitutional and legal tensions surrounding dual-use space infrastructure in EU governance. Building on analysis of Article 189 TFEU and CJEU case-law, the article argues that existing EU legal interpretations render nearly all dual-use assets as "civilian", thereby extending EU competence in ways that raise Member State sensitivities. This absolutist reading, the author contends, is ill-suited to the operational and political realities of dual-use governance. Drawing from international legal debates and disarmament regimes, Bennett proposes a more nuanced tiered model for classifying dual-use space systems within EU law. The article is a key contribution to debates over how the EU can act strategically while respecting constitutional constraints.

2.3 | Cybersecurity and the EU Space Law (Stefoudi)

In her contribution, Stefoudi examines the cyber vulnerabilities of satellite infrastructure and evaluates how existing EU cybersecurity law—notably the NIS 2 and CER Directives—falls short of comprehensively addressing threats to space-based systems. The article explores how a forthcoming EU Space Law could fill these gaps by establishing common cybersecurity standards and enhancing regulatory coverage of space-to-space and space-to-ground vulnerabilities. In doing so, Stefoudi identifies a major blind spot in current EU policy-making and provides actionable legal pathways to close it. The article is timely, given the rise of cyber-physical threats in space and the blurred boundaries between civil, commercial, and defense infrastructure.

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2.4 | Planetary Boundaries and Environmental due Diligence (Cirkovic and Braun)

The article by Cirkovic and Braun introduces an ecological perspective to space governance, focusing on the environmental externalities of satellite constellations and launch activities. Cirkovic and Braun argue that the EU Space Law should integrate environmental impact assessments (EIA) and corporate due diligence frameworks, drawing on the Corporate Sustainability Due Diligence Directive and existing EU environmental law. The authors position the EU as a potential leader in embedding environmental responsibility into space law, particularly through lifecycle approaches and intergenerational equity principles. This article uniquely bridges Earth system science, space law, and sustainability governance, and calls for legal innovation at the intersection of terrestrial and extra-terrestrial stewardship.

2.5 | Targeting Non-EU Operators Under EU Space Law (Baumann and Prall)

Baumann and Prall critically assess the extraterritorial dimensions of the EU's planned Space Law. Their analysis interrogates how rules intended to ensure safety, security, and sustainability in space could apply to non-EU actors operating in European markets. They examine the legal implications of defining "services provided in the EU" and the risk of regulatory overreach. The article also discusses tensions with international trade law and space law principles. By surfacing the risks of unintended market barriers, the authors stress the need for careful legal engineering if the EU is to be a rule-maker rather than a market-closer in space.

2.6 | Ariane, Integration, and Europe's Launcher Crisis (Popper and Klimburg-Witjes)

This interdisciplinary article applies Science and Technology Studies (STS) perspectives to the Ariane launcher program. It explores how Ariane has historically embodied European integration but is now at the center of a legitimacy and functionality crisis. The delays surrounding Ariane 6 and Europe's increasing reliance on SpaceX reflect a deeper governance crisis. The authors contrast monocentric and polycentric visions of future launcher development, and interpret Europe's current crossroads as a reflection of broader tensions between supranationalism, national sovereignty, and private-sector agility. The article is an original intervention in the politics of European space infrastructure and industrial capacity.

2.7 | Rethinking Industrial Policy for EU Space (Dalledonne and Dousset)

The final article for this special issue turns to the challenge of industrial competitiveness and calls for a unified EU-wide space industrial policy. The authors dissect the fragmented "governance triangle" of the EU, ESA, and Member States, and argue for harmonized procurement, strategic investment in critical technologies, and reciprocal market access policies. Their analysis connects industrial fragmentation to broader risks in Europe's strategic autonomy and economic resilience. The article offers concrete recommendations for a coordinated industrial policy that aligns with Europe's defense and sustainability goals, echoing themes raised in several other contributions.

2.8 | Between Ambition and Restraint

Taken together, the articles in this special issue explore the legal, institutional, ecological, and industrial dimensions of a European space sector in transition. They reveal the multifaceted nature of EU space governance, where constitutional complexity, regulatory ambition, and geopolitical urgency intersect. Rather than presenting a linear narrative of integration, these contributions illuminate a landscape marked by contestation, experimentation, and adaptation.

A common thread running through all seven articles is the tension between ambition and constraint: between the Union's aspirations to be a strategic actor in space and the limitations of its legal architecture, institutional fragmentation, and uneven industrial capacities. Several articles also point to the risks of overreach—legal, political, or environmental—and emphasize the need for proportionate, carefully engineered, and balanced governance tools.

As such, this special issue provides both a diagnostic and a roadmap. It demonstrates that if the EU is to succeed in its ambitions for space, it must grapple with the fundamental design questions of its governance model: Who decides? Who benefits? And who is responsible? By foregrounding these questions, the articles in this issue contribute not only to scholarly debate but also to the policy agenda shaping Europe's future beyond Earth.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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